

CS383

Software Requirements Specification (SRS)

Barakah App

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1. Introduction

This document provides a detailed overview of the requirements for developing a food donation application. It aims to reduce food waste and efficiently connect donors with charitable organizations. The key features of the application include account creation, donation request management, and communication tools between relevant parties, in addition to search and filtering functionalities for charitable organizations. The project covers functional requirements, which outline the core tasks of the application, such as user registration, request processing, and data management, as well as non-functional requirements focusing on usability, high performance, digital security, and compatibility with various devices and operating systems. The application ensures a seamless user experience through an intuitive interface and integration with external systems, such as databases and notification services, to enhance coordination and communication efficiency.

2. Requirements engineering

Requirements engineering is the process of identifying, gathering, and defining the system's required functionalities to meet user needs.

2.1.1 Requirements Discovery

This step aims to understand the objectives and functional needs of the food preservation application. It is achieved using various techniques like interviews, surveys, and data analysis. The key system requirements include:

- Users should be able to sign up by providing contact information, role selection, and additional details.
- Users must be able to log in, with password verification and error handling.
- The system should provide data management functionalities, allowing users to add, delete, and modify their data related to their personal profile..
- The system should allow users to view a list of charities.
- The system automatically adds charities from an external reliable source.
- Users should be able to submit a preservation request, which includes donor information, food details, donation scheduling, location, validation, and initial confirmation.
- Once the user confirms the request, the system automatically sends it to the administrator for further processing.
- The administrator is responsible for managing preservation requests. They can review each request, modify its details if needed, approve or reject it, and assign a volunteer to deliver the request to the designated charity.
- Users should have access to communication and coordination tools, including messaging, notifications, requesting assistance, pickup and delivery operations, monitoring and coordinating tasks.

- The administrator can perform monitoring and coordination. Additionally, they should manage task assignment and conflict resolution to ensure smooth donation and delivery processes.
- The system should allow users to switch accounts when needed by logging out and signing in with different credentials.
- A database should be integrated to store and manage system data securely.

2.1.2 Requirements Prioritization

The requirements are prioritized based on their importance, urgency, and feasibility. The prioritization criteria include:

- **Criticality:** Identifying essential functions such as authentication, request handling, and data management, as they form the core functionality of the system.
- **User Impact:** Focusing on features that enhance the overall experience, ensuring intuitive navigation, effective communication, and seamless interaction.
- **Technical Feasibility:** Evaluating the complexity and practicality of each requirement to prioritize those that provide significant value while maintaining implementation efficiency.

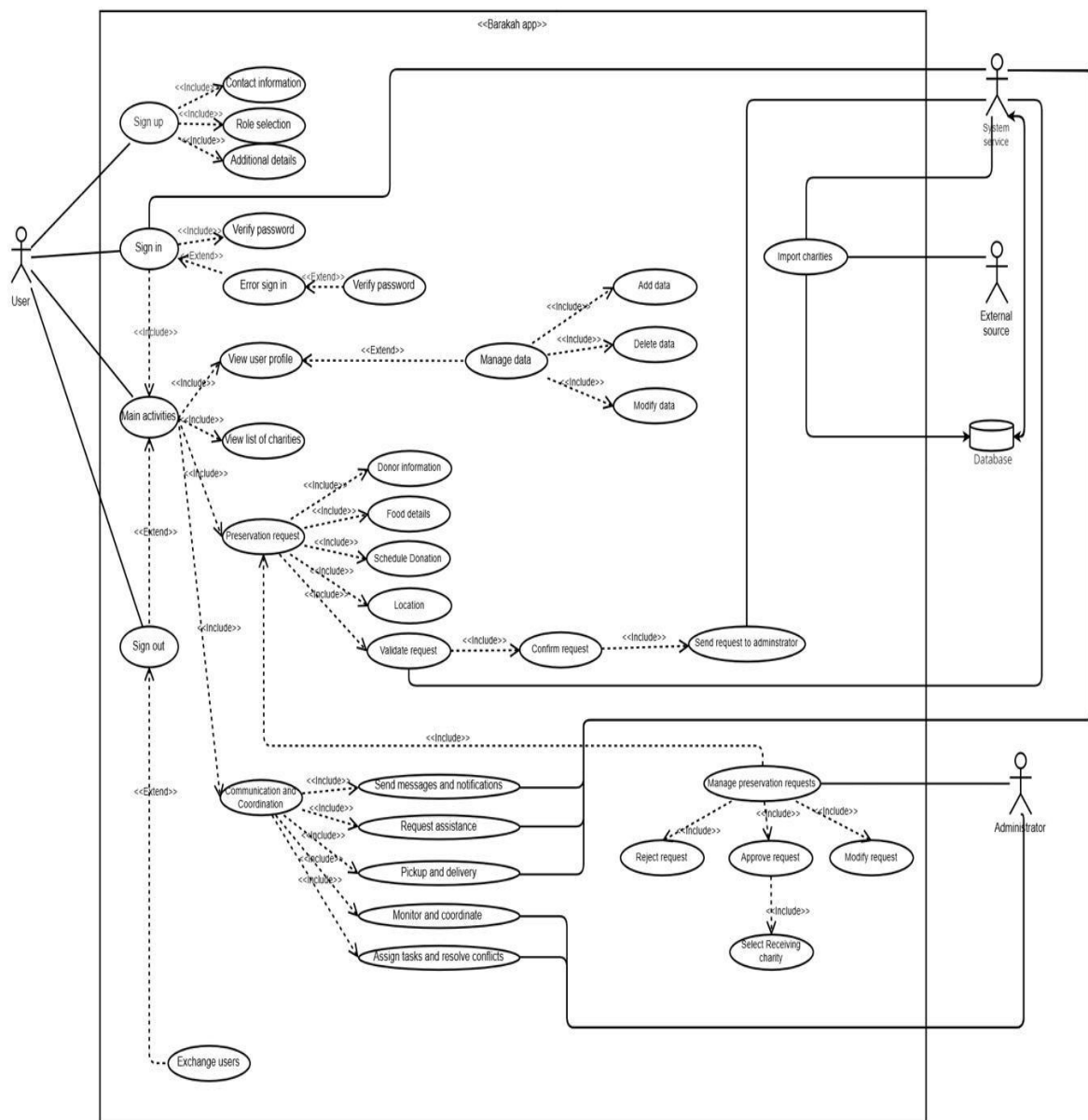
2.1.3 Requirements Specification

The prioritized requirements are documented clearly to guide development and testing. This includes:

- Defining data input and output formats for managing donations and food preservation requests (donor information, food details, scheduling,).
- Specifying user interface interactions for submitting and tracking through monitor requests related to food preservation.
- Ensuring that users can modify their requests before confirming, while only the administrator has the authority to modify or cancel a request after approval.
- Ensuring security and performance requirements for handling user data and system transactions efficiently and securely in a database.
- Making requirements measurable, testable, and unambiguous to facilitate effective development, such as confirming successful account creation, request approval, and notification delivery.

2.2 Requirements Collection

After analyzing the system requirements, they are illustrated through the Use Case Diagram to visualize the system's core functionalities, user interactions, and administrative controls.



2.3 Requirements Validation

- We have checked that the donor and volunteer can log in when they enter the correct data, the donor shows them the powers of managing their donations, and the volunteer the powers of managing requests for aid or delivery.
- We checked that the app refuses to log in when entering incorrect data, showing the message 'unregistered email' if the account does not exist, and the message 'incorrect password' when entering the wrong password.
- We checked that when entering correct data, the donation should be saved and added to the list of available donations.
- We checked when the data was correctly modified, the modifications and donation update should be saved in the list.
- We checked when you deleted the donation should not appear on the list after the update.
- We checked that the application appears to the donor once submitted by the beneficiary.
- We checked that all users' information is encrypted and protected from unauthorized access.
- We checked that the app works efficiently when there are a large number of requests. The results appear in less than 3 seconds.
- We checked the geographical locations of the beneficiary and donor, and the entrance location is correct and accessible.
- We checked that the app interface is easy to use for donors, making the donation process more convenient and easier.
- We checked the backup storage securely.
- We checked that food images are supported with high quality and safety.
- We checked that the electronic payment process is secure and series, with data protection and easy tracking of transactions.

3. Functional Requirements

3.1 User Management

The system allow Users to register by entering their contact information, selecting their role (Donor, Volunteer) and providing additional details based on their role.

They can use sign up if they new to the system, or sign in if they already have an account.

- **Sign Up**

Input:

1. User full name
2. Username
3. Phone number
4. Email address
5. Password
6. Role(Donor, Volunteer)

Output:

- 1- user would be converted to the home page.
- 2- varivacation message is sent to the email address to ensure security and validation.
- 3-user account is created.

- **Sign in**

Input:

1. Email address or Username
2. Password

Output:

1. User data is verified.
2. User directed to the activities can be done by their role.

- **Sign out**

Input:

1. Switching Account or Delete Account

Output:

If the user select Switching Account he will be able to get into the other Account and benefit from the activities based on the role, But if he selected Delete Account the Account information will permanently Deleted from Data Base unless he wants to delete it without deleting information from DB.

3.2 Search and filtering

The system would allow Users to view a list of available charities. User would be able to filter and search for charities and search for a specific name.

Input:

1. Search by Charitie name.
2. Filters include (location, time).

Output:

1. System would generate available list of charities based on user search.
2. Listed charities would include details like name, location, type services provide.

3.3 Donation And Request Management

The system would allow a donor to create a donation request Once submitted, the system would store the request and allow an administrator to review, approve, modify, or reject it. Upon approval, the Administrator shall assign the request to an available charity for distribution.

Input:

1. Donor information
2. Food Details
3. Location
4. Schedule donation

Output:

1. Request is validated and confirmed.
2. Confirmation Message :The donor receives a notification about request submission.
3. Approval/Rejection Notification : The donor is informed about the request status.
4. Task Assignment Alert : An available charity is notified when a request is assigned to them.

3.4 Communication And Alignment

Users can communicate between each other like a Donor and Volunteer. The system shall allow administrators to assign a preservation task to a user with the volunteer role after a donation request has been approved. Once assigned, the system shall send alerts to both the donor and the volunteer, notifying them of their assigned task and enabling communication.

Input:

1. Approved preservation request (from the donor).
2. Administrator selects a volunteer for the task.

Output:

1. Task assigned notification to the volunteer with donor details and preservation request information.
2. Task confirmation notification to the donor with volunteer details.
3. Messaging system activated between donor and volunteer for coordination.

3.5 Data Management & Administration

The system allows administrators to manage user data, including editing, deleting, or deactivating accounts when necessary. Additionally, all transactions, requests, and communications are securely stored for tracking and auditing purposes

Input:

1. User data (name, email, phone number, role)
2. Type of action (edit, delete, deactivate)
3. Transaction, request, or communication details

Output:

1. Successful update, modification, or deactivation of user accounts
2. All modifications and actions logged in the database
3. Ability to review stored records for auditing and security purposes

3.6 Reporting and Analytics

The system allows administrators and authorized users to generate reports, track requests, and analyze platform activity to improve operational efficiency and decision-making

Input:

1. User role (Admin, Supervisor)
2. Request type (Donation, Assistance)
3. Date range selection

4. Request status filter (Approved, Rejected, Pending)

Output:

1. A list of filtered requests with status and details
2. A statistical report on user activity and request trends
3. Graphical representation of donation and assistance patterns

3.7 Notifications and Alerts

The system provides real-time notifications and alerts to users regarding their requests, messages, and system updates.

Input:

1. User account credentials
2. Request status updates
3. New messages received

Output:

1. Error message on failed login attempts
2. Notification for request approval, modification, or rejection
3. Instant alert for new messages from charities or administrators

3.8 User Profile Management

The system allows users to manage their personal information, update profile details, and switch between accounts if needed.

Input:

1. Full name
2. Username
3. Email address
4. Phone number
5. Password
6. Role (Donor, Charity Organization)

Output:

1. Profile successfully created or updated
2. If the role is a charity organization, additional verification is required (Organization name, location, phone number, website)
3. Verification message sent to the email for security validation
4. User can switch between accounts without logging out

4. Non-Functional Requirements

4.1 Usability

The application provides a user-friendly experience by ensuring an intuitive interface for donors, volunteers, and administrators. Users can easily navigate through the system to sign up, manage donation requests. The app includes a smart filtering system to help volunteers and organizations find relevant food donation requests based on location, type of food, and urgency. The system ensures a smooth and responsive design, allowing users to access it seamlessly across different devices, including desktops, tablets, and smartphones.

4.2 Performance

The application is capable of processing a high volume of donation requests and user actions without noticeable performance issues. It ensures fast response times, with all critical actions such as logging in, submitting a donation request, retrieving data, and updates order statuses. The system supports multiple concurrent users without slowdowns, ensuring seamless operation even during peak usage. Additionally, the database is optimized for quick searching, retrieval, and validation of donation requests, maintaining high performance as the platform scales.

4.3 Security

The system applies secure authentication and authorization procedures to ensure that only valid users have access to manage or approve donation requests. User credentials are securely stored using encryption techniques, and two-factor authentication (2FA) is enforced for added security. Also provides role-based access control to prevent unauthorized modifications to donation records. The platform protects all sensitive data, including personal user details and donation records, by encrypting them both in transit and at rest. Regular security audits and vulnerability assessments are conducted to identify and address potential risks.

4.4 Maintainability

The application is built using a modular architecture and well-documented code structure, ensuring ease of updates and enhancements without disrupting core functionalities. It follows industry best practices for clean and structured coding, making debugging and feature expansion more efficient. Logs are stored to track errors and system activities, ensuring efficient issue resolution.

4.5 Compatibility

The system is designed to be fully compatible with the latest versions of major web browsers, including Chrome, Firefox, Safari, and Edge. It supports cross platform functionality, allowing users to access it on Windows, macOS, iOS, and Android devices. The responsive design ensures optimal performance across different screen sizes, maintaining usability on desktops, tablets, and mobile phones.

4.6 Scalability

The platform is designed to support increasing numbers of users and donation requests without major architectural changes. It employs scalable infrastructure, ensuring smooth operation even as demand grows. The database is structured to handle a growing volume of data efficiently.

5. External interface requirements

5.1 User Interface

- **Description:** The app features a simple, user-friendly interface for managing donations efficiently.
- **Functionality:** Users can register and log in to create or manage donation requests. The interface allows donors to upload images of food donations and specify details such as quantity, type, and pickup location. Volunteers and charity organizations can browse and claim donations based on availability. The app supports notifications for request updates, status changes, and communication between users.
- **Technical Aspect:** A responsive design ensures compatibility across smartphones, tablets, and desktops. The app ensures smooth performance with optimized loading times and minimal clutter.

5.2 Hardware Interface

- **Description:** Compatible with smartphones, tablets, and computers.
- **Functionality:** The app supports camera access for scanning food images and uploading donation details. GPS functionality helps users set accurate pickup locations. Supports cloud storage integration for secure data storage and retrieval.
- **Technical Aspect:** Supports cloud storage integration for secure data storage and retrieval. Ensures compatibility with different operating systems (iOS, Android, Windows).

5.3 Messaging & Notifications

- **Description:** Integrates with WhatsApp and Telegram for communication.
- **Functionality:** Users can receive real-time notifications regarding donation updates, volunteer assignments, and charity confirmations. Donors, volunteers, and charities can communicate directly through secure messaging.
- **Technical Aspect:** Uses APIs such as WhatsApp Business API and Telegram Bot API to facilitate real-time messaging.

5.4 Security & Privacy

- **Description:** The system ensures user data privacy and security by implementing encryption and authentication mechanisms.
- **Functionality:** Secure user authentication through password encryption and optional two-factor authentication. Data protection measures to prevent unauthorized access.
- **Technical Aspect:** Uses end-to-end encryption for sensitive data. Compliance with data privacy regulations to safeguard user information.

